



The evolution of design

Tatjana Dzambazova, AI Strategy, Office of the CTO, Autodesk on how the use of AI in design has led to a revolutionary change.

Mithun Mohandas | mithun@digit.in

d Could you elaborate a little about Generative Design?

Tatjana: We're on the verge of a revolution and it's called infinite computing. It's happening because of three things. 1. There's an exponential rise of computing power. 2. Democratised access to that power and 3. The precipitous fall of the cost to the power. Combine that with the cloud's ability to scale and you're looking at the possibility of applying hundreds of thousands of computers to solve problems. With infinite computing, you can solve complex problems using infinite number of resources to solve a problem at no additional cost. And it will change our lives and the way we design.

Previously, when we designed, we'd have to direct the tool to tell it what to do and then we'd simulate, and then simulate the design to look for defects. This iterative process where you'd get the results of the simulation after a couple of hours took a lot for time and when the results weren't desirable, you'd have to go back to the drawing board. It's not a scalable model. You might have arrived at a better design, but not necessarily an optimal design. With the power of the cloud you can mesh and extrapolate many variations and tell you the trade-offs. That's what we call Generative design. It's a methodology that asks the designer to define just the goals. The goals can be cost, materials, weight or

any criteria that describes the problem. And then using smart algorithms that are usually based on evolutionary patterns, it generates many options within those design specs. We can then check all the options, understand the trade-offs and continue to choose the proper design.

d So the methods are going to be used to improve designs?

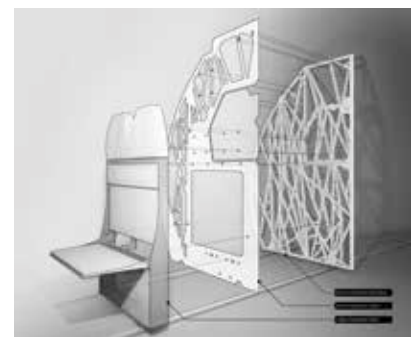
Tatjana: We'll not be applying simulations on top of already existing designs like we have until now, or even during the design process. Instead, we will use simulation to discover design, to discover all the possibilities within the environment, that we as designers, define. So this will result in obviously faster time to production and lesser cost. It's not just the functional aspect, but also allows us to design in a new way and generate forms that we would never have initially thought of. It's augmenting our creativity.

d You mentioned that these designs were based on evolutionary patterns, could you give us an example?

Tatjana: Take the example of Lightning Motorcycle, the fastest electric motorcycle in the world. They continuously look at how to improve the design. The swing arm of the bike is super heavy since it holds the weight of the entire bike but they wanted to make it lighter so that the bike runs even faster. So they ran it through a generative design tool and the tool started to reduce and add material, trying out different approaches and came

up with a new design which looks very similar to a cat's pelvis. It tries to mimic nature. The fact is that nature always uses the minimum amount of energy necessary for maximum performance. This is what generative design does. In this case, we put the form of the swing arm and asked for it to be optimised. However, true generative design is where you give the constraints i.e. points/lines of action for the force, weight, wind, etc. And the system generates a design.

Another example is that of Airbus. They want to make flying more experimental. They have these partitions within



aircrafts which are very heavy because the stewardess' seats were attached on it. So they ran it through an algorithm based on slime mould patterns. Slime mould produces roots of branches which look for food, the roots which find food get to survive and those which don't die. So they ended up with a design that uses 1/3rd of the material, is 45 percent lighter and is much sturdier.

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